

1 Pre-assembly Preparation

1.1 Overview

Before assembling the robot arm, we need to adjust all servo motors to 90 degrees so that they can be mounted at the neutral position of the mechanical structure and ensure the robot arm has sufficient degrees of freedom.

1.2 Required Components

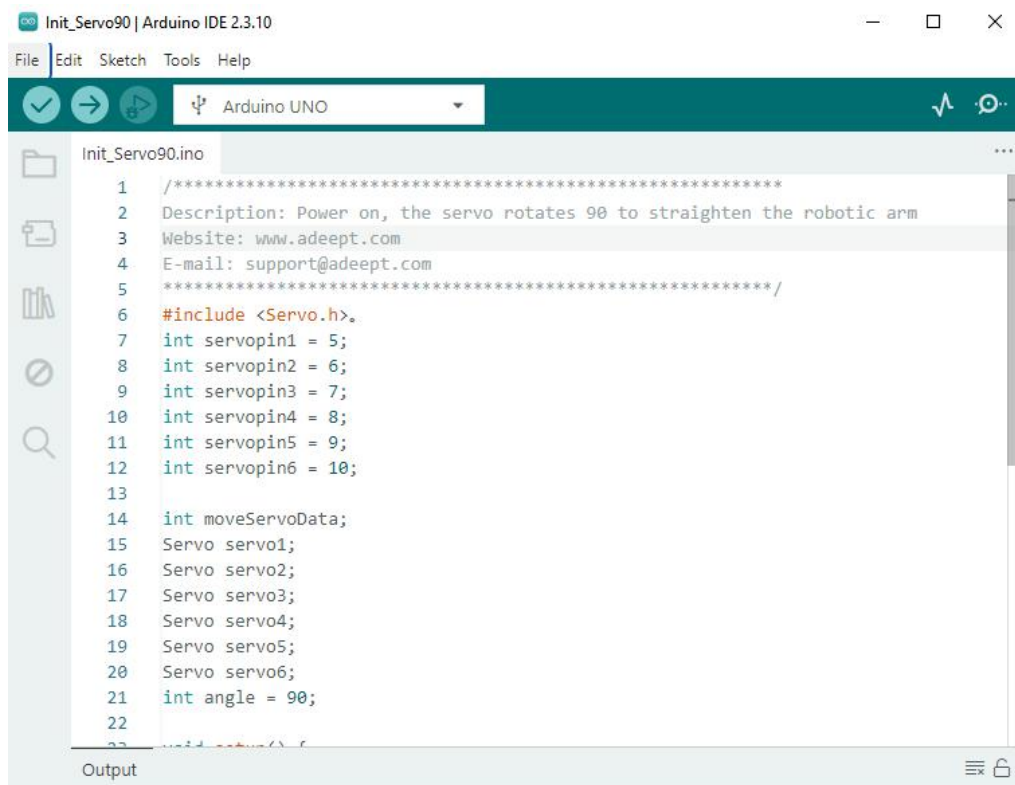
Components	Quantity	Picture
Adeept Pixie Board	1	
Type-C USB Cable	1	
Servo	5	

1.3 Wiring Diagram

Connect the servo to ports S6-S2, Pay attention to the pin correspondence.

1.4 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board with a USB cable. (Servos require a large operating current, so power the board with batteries and turn on the power switch.)
2. Open "Init_Servo90" folder in "03 Robotic Arm Control/Code" , double-click "Init_Servo90.ino" .



```
Init_Servo90 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
[Check] [Run] [Serial] [USB] Arduino UNO
Init_Servo90.ino
1  /*****
2  Description: Power on, the servo rotates 90 to straighten the robotic arm
3  Website: www.adeept.com
4  E-mail: support@adeept.com
5  *****/
6  #include <Servo.h>
7  int servopin1 = 5;
8  int servopin2 = 6;
9  int servopin3 = 7;
10 int servopin4 = 8;
11 int servopin5 = 9;
12 int servopin6 = 10;
13
14 int moveServoData;
15 Servo servo1;
16 Servo servo2;
17 Servo servo3;
18 Servo servo4;
19 Servo servo5;
20 Servo servo6;
21 int angle = 90;
22
Output
```

3. Select development board and serial port.

Board: **Tools---**>**Board---**>**Arduino AVR Boards---**>**Arduino Uno**

Port: **Tools --->Port---**>**COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```
Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.
```

5. After the program runs successfully, all servo will rotate to 90 degrees.

1.5 Code

```
01  /*****
02  Description: Power on, the servo rotates 90 to straighten the robotic arm
03  Website: www.adeept.com
04  E-mail: support@adeept.com
05  *****/
06  #include <Servo.h>
07  int servopin1 = 5;
08  int servopin2 = 6;
09  int servopin3 = 7;
10  int servopin4 = 8;
11  int servopin5 = 9;
12  int servopin6 = 10;
13
14  Servo servo1;
15  Servo servo2;
16  Servo servo3;
17  Servo servo4;
18  Servo servo5;
19  Servo servo6;
20  int angle = 90;
21
22  void setup() {
23    servo1.attach(servopin1);
24    servo2.attach(servopin2);
25    servo3.attach(servopin3);
26    servo4.attach(servopin4);
27    servo5.attach(servopin5);
28    servo6.attach(servopin6);
29
30    servo1.write(angle);
31    servo2.write(angle);
32    servo3.write(angle);
33    servo4.write(angle);
34    servo5.write(angle);
35    servo6.write(angle);
```

```
36     delay(20);  
37 }  
38  
39 void loop()  
40 {  
41  
42 }  
43
```